(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendurthy (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.VIROTI HARISH KUMAR**

Registration Number: **324129520049**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

SMART HOME ELECTRICITY CONSUMPTION PREDICTION SYSTEM USING HISTORICAL ENERGY USAGE PATTERNS

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mrs.S.SIRISHA

*Department of **Mechanical Engineering***

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr.VIROTI HARISH KUMAR

Reg.No: 324129520049

*Department of **Mechanical Engineering***

Department of Mechanical Engineering

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever nssary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I, Mr.**VIROTI HARISH KUMAR**, a student of **Bachelor of Technology** Program, Reg. No. **324129520049** of the Department of **Mechanical Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mrs.S.SIRISHA**, Department of **Mechanical**, **Wellfare Institute of Science, Technology and Management**.

V. Harish Kumar

(Signature and Date)

Official Certification

This is to certify that Mr.**VIROTI HARISH KUMAR** , Reg. No. **324129520049** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **SMART HOME ELECTRICITY CONSUMPTION PREDICTION SYSTEM USING HISTORICAL ENERGY USAGE PATTERNS** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Mechanical Engineering** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsement

Faculty Guide

Head of the Department,
Department of Mechanical Engineering
Wellfare Institute of Science
Technology and Management
Pinagadi, Visakhapatnam
Andhra Pradesh-531173

Head of the Department

Principal

Certificate from Intern Organization

This is to certify that Mr.**VIROTI HARISH KUMAR**, Reg. No. **324129520049** of **Wellfare Institute of Science, Technology and Management**, underwent internship in **SMART HOME ELECTRICITY CONSUMPTION PREDICTION SYSTEM USING HISTORICAL ENERGY USAGE PATTERNS And Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

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Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr.R.Sagar**, Head of the Department of **Mechanical Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mrs.S.SIRISHA**, Assistant Professor of the Department of **Mechanical Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing a Smart Home Electricity Consumption Prediction System using Historical Energy Usage Patterns. The internship spanned an 8-week period and was undertaken to apply advanced machine learning and predictive analytics methodologies to the energy management sector, specifically addressing the critical need for automated electricity forecasting. The primary objective of this internship was to gain proficiency in time-series data analysis, regression modeling, and energy informatics to enhance employability skills while solving a major sustainability challenge.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a machine learning regression system using Python and Scikit-learn that can accurately forecast future household electricity consumption based on historical usage patterns.

- To understand and process time-series energy datasets, specifically focusing on extracting and utilizing key indicators such as seasonal variations, temperature correlations, HVAC usage, and occupancy levels.

- To evaluate and compare different regression architectures, including Linear Regression, Random Forest Regressor, and Gradient Boosting, to determine the most effective algorithm for high-dimensional smart home data.

- To implement robust evaluation metrics suitable for regression problems, including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2) score, understanding how to minimize prediction variance.

• To design an automated analytical pipeline that generates interpretable energy insights, allowing homeowners and utility providers to understand the specific factors driving electricity demand.

1.2 Outcomes Achieved

Key outcomes from my internship include:

• A fully operational predictive regression engine capable of forecasting daily electricity consumption, utilizing an ensemble Random Forest model on a complex, multi-variable smart home dataset.

• Energy management companies and utility providers can utilize this automated forecasting logic as a backend planning tool, significantly improving grid load balancing and dynamic pricing strategies.

• Comprehensive data visualizations including consumption distributions, model performance comparisons, temperature correlation plots, and feature importance rankings that enhance the interpretability of the machine learning models for homeowners.

• A robust feature engineering pipeline that successfully quantifies the compound effect of environmental factors (temperature, humidity) and behavioral factors (occupancy, appliance usage) alongside traditional historical metrics.

• The prediction system establishes a foundation that can be extended with deep learning architectures (like LSTMs processing hourly smart meter data) for real-time energy load forecasting.

These outcomes directly address the problem statement by providing a modern and intelligent energy prediction solution that improves electricity consumption forecasting, promotes efficient energy utilization, reduces operational costs, and supports sustainable energy management.

The successful implementation of this system demonstrates the powerful intersection of artificial intelligence and sustainable energy management. By moving away from purely reactive monitoring methods and toward automated, machine learning-driven predictive assessment, homeowners can proactively manage their energy footprint, potentially revolutionizing how communities interact with the power grid.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization hosting this internship is a leading clean-tech solutions provider focused on bridging the gap between smart grid research and practical data science, enhancing student employability, and promoting innovation in the sustainable energy sector. By leveraging emerging technologies such as Machine Learning and IoT Analytics, the organization aims to augment the smart home ecosystem, enabling utility providers and consumers to utilize their usage data for proactive energy management.

The organization's collaborations with prominent energy institutions and smart device manufacturers underscore its value and credibility in the sustainability sector. Through projects like the Smart Home Electricity Consumption Prediction System, the organization demonstrates its commitment to applying cutting-edge AI to solve pressing environmental challenges, specifically within the realm of automated energy forecasting.

2.2 Vision, Mission, and Values

- **Vision:** To combine cutting-edge data science with impactful IoT solutions to drive sustainable innovation and automated energy management in global smart grids.

- **Mission:** To support organizations dedicated to clean energy by empowering and equipping professionals with intelligent predictive tools, thereby creating a highly efficient, data-driven utility ecosystem.

- **Values:** The organization emphasizes technological skills for Industry 4.0, algorithmic accuracy, strict consumer data privacy, and ethical AI development for a sustainable future.

2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the organization's mission. Interns are expected to adhere to the following policies:

- **Confidentiality and IP Compliance:** Interns must maintain the strict confidentiality of all organizational and proprietary smart meter data, adhering to consumer privacy standards.

- **Professionalism:** Interns are expected to demonstrate professionalism, punctuality, and respect for all team members, energy engineers, and mentors.

- **Learning and Contribution:** Interns are encouraged to actively participate in projects, share innovative ideas regarding machine learning applications in sustainability, and contribute to the organization's goals.

- **Compliance:** Interns must comply with all organizational policies, including ethical guidelines for IoT AI development and software engineering best practices.

2.4 Organizational Structure

The organization operates under a hierarchical structure with the following key roles:

- **Board of Directors:** Provides strategic direction and oversight for Clean-Tech and Data Science initiatives.

- **Executive Director:** Oversees day-to-day operations and implementation of energy analytics programs.

- **Project Managers:** Lead specific initiatives such as the development of forecasting software and smart grid AI tools.

- **Data Science Team:** Conducts research, develops machine learning models, and engages in technical innovation for sustainable energy.

• **IoT Engineering Board:** Provides hardware expertise to ensure algorithms align with current smart meter telemetry standards.

• **Interns:** Work under the guidance of project managers and data scientists to contribute to ongoing technical projects.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are typically placed under the guidance of project managers or data science teams. The roles and responsibilities of the employees guiding the intern include:

1. Project Managers:

• Design and implement technical clean-tech projects.

• Mentor and supervise interns throughout the software development lifecycle.

• Coordinate with utility stakeholders to gather forecasting requirements for predictive tools.

2. Data Scientists:

• Conduct research on regression algorithms for energy demand forecasting.

• Prepare complex time-series datasets and feature engineering pipelines.

• Analyze data and provide technical recommendations for model optimization to minimize prediction variance.

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Interns assist these teams by conducting research, drafting technical documents, developing Python code, and supporting IoT data analysis efforts. The collaborative environment ensures that interns receive comprehensive training in both theoretical concepts and practical implementation of AI systems within the highly dynamic context of energy informatics.

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CHAPTER 3

PROBLEM ASSESSMENT

3.1 Problem Analysis

Efficient energy management has become increasingly important due to rising electricity demand and growing environmental concerns. Traditional methods of monitoring household electricity usage provide limited insights into future consumption, making it difficult for homeowners to optimize energy utilization and reduce costs. Intelligent prediction systems are required to analyze historical energy usage and forecast future electricity consumption accurately.

Traditional energy monitoring methods face several systemic limitations:

- Reactive Analysis: Most current smart home dashboards only show historical usage, providing no actionable foresight for the upcoming days or weeks.
- Failure to Account for Complex Variables: Simple historical averaging fails to account for the non-linear impact of sudden weather changes, shifting occupancy, or weekend behavioral patterns.
- Peak Load Blindness: Without accurate forecasting, utility companies struggle to predict localized peak demand, leading to inefficient grid balancing and higher costs passed to consumers.
- Lack of Granular Optimization: Homeowners cannot optimize heavy appliance usage (like EV charging or laundry) to off-peak times if they do not have a reliable forecast of their baseline demand.

Consequently, the energy sector requires intelligent, automated systems that can analyze raw telemetry data, extract behavioral and environmental parameters, and apply machine learning to forecast demand objectively and instantaneously.

3.2 Key Parameters

To accurately predict electricity consumption, the machine learning system must evaluate several intersecting environmental, behavioral, and historical parameters extracted from smart home data:

- Temporal Features: Day of the Week, Month, Day of the Year, and specific flags for Weekends and Holidays, which strongly dictate behavioral routines.
- Environmental Metrics: External Temperature and Humidity, which are the primary drivers of heavy HVAC (Heating, Ventilation, and Air Conditioning) utilization.
- Behavioral Indicators: Occupancy Level (number of people in the home) and Appliance Usage Count (frequency of major appliance activation).
- System Metrics: HVAC Usage Hours, representing the most significant continuous power draw in a modern smart home.
- Historical Baseline: Previous Day Consumption, providing the autoregressive foundation for the time-series prediction.

3.3 Requirements Evaluation

The proposed Smart Home Electricity Consumption Prediction System must meet specific functional and non-functional requirements to address the energy management challenge effectively.

Functional Requirements:

- Data Integration: The system must accept and process standard time-series features extracted from smart meter APIs.
- Regression Forecasting Engine: The system must utilize trained machine learning models to predict continuous numeric values representing future kWh consumption.

• Energy Reporting: The system must generate detailed analytical reports breaking down consumption by temperature ranges, occupancy levels, and days of the week.

• Individual Assessment: The system must output specific daily forecasts with confidence intervals to indicate the certainty of the algorithmic prediction.

• Visualization: The system must generate visual dashboards illustrating the distribution of key energy parameters and the correlation between temperature and consumption.

Non-Functional Requirements:

• High Accuracy: The system must achieve a low Root Mean Squared Error (RMSE) to ensure the forecasts are financially useful for cost-saving decisions.

• Interpretability: The system should provide clear explanations (feature importance) for what environmental factors are driving the demand forecast.

• Robustness: The models must maintain accuracy across diverse seasonal changes and handle sudden behavioral anomalies (e.g., unexpected holidays).

• Scalability: The architecture must be capable of processing large batches of smart home metadata for neighborhood-level forecasting.

By mapping these requirements directly to the identified energy parameters, the solution ensures a comprehensive and effective approach to modern smart grid management. The reliance on data-driven regression algorithms directly addresses the limitations of reactive, historical-only monitoring.

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CHAPTER 4

SOLUTION DESIGN

4.1 Solution Blueprint

The Smart Home Electricity Consumption Prediction System is designed as a modular machine learning pipeline that transforms complex time-series telemetry data into accurate continuous forecasts and actionable energy insights. The blueprint consists of three primary components:

1. IoT Data Generation and Preprocessing Module:

This component is responsible for creating a robust, realistic dataset that mimics actual smart meter extractions over a two-year period (730 days). It includes a synthetic data generator that models complex environmental relationships: e.g., applying sinusoidal waves to simulate seasonal temperature variations, and applying logical multipliers to simulate increased weekend occupancy. The preprocessing pipeline utilizes Scikit-learn's StandardScaler to normalize continuous variables (like Temperature and Previous Consumption), ensuring optimal performance for gradient-based algorithms.

2. Predictive Modeling Engine:

This is the core analytical component. It implements three distinct regression algorithms to predict the continuous target variable (Electricity Consumption in kWh):

- Linear Regression: A fundamental statistical model that establishes a baseline. It calculates the predicted consumption based on a linear combination of the environmental and behavioral features.

- Random Forest Regressor: An ensemble learning method that constructs multiple decision trees. It is highly effective at capturing complex, non-linear interactions between energy parameters (e.g., the combined effect of high temperature and high occupancy on a weekend), making it excellent for nuanced demand forecasting.

• Gradient Boosting Regressor: A sequential ensemble technique that builds trees to correct the residual errors of previous trees, optimizing for the lowest possible variance in predicting fluctuating power loads.

3. Analytics and Visualization Dashboard:

This component translates the mathematical outputs of the models into interpretable sustainability intelligence. It generates comprehensive visual reports, including parameter distributions, model performance comparisons (RMSE, MAE, R^2), prediction vs. actual scatter plots to analyze forecasting accuracy, and feature importance rankings.

4.2 Feasibility Assessment

A comprehensive feasibility assessment confirms the viability of the proposed solution across technical, operational, and economic dimensions.

Technical Feasibility: The project is highly technically feasible. It leverages the mature Python data science ecosystem, specifically Pandas for time-series manipulation and Scikit-learn for machine learning. These libraries provide robust, optimized implementations of the required regression algorithms, ensuring that the system can handle the high-dimensional complexity of smart meter data.

Operational Feasibility: The system is operationally feasible as it addresses a universal need in smart home management. The automated nature of the prediction engine means it can be integrated into existing Home Assistant platforms or utility provider apps, acting as an automated forecasting layer that updates daily.

Economic Feasibility: The project is economically sound. By utilizing open-source Python libraries, the development avoids expensive proprietary forecasting software licenses. Furthermore, the system provides massive economic value to consumers; automating the forecasting process allows homeowners to shift heavy loads to off-peak pricing hours, directly reducing monthly utility bills.

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4.3 Implementation Plan

The development of the Smart Home Electricity Consumption Prediction System followed a structured implementation plan, divided into four key phases:

• Phase 1: Requirement Analysis and Data Engineering (Weeks 1-2): Defined the key environmental parameters influencing energy consumption based on smart grid literature. Developed the synthetic data generation script to create a realistic dataset of 730 daily records incorporating seasonal, behavioral, and historical data.

• Phase 2: Model Development and Training (Weeks 3-4): Implemented the preprocessing pipeline using standard scaling. Developed, trained, and tuned the Linear Regression, Random Forest, and Gradient Boosting models using an 80/20 train-test split methodology, preserving the temporal integrity of the data.

• Phase 3: Analytics and Visualization Implementation (Weeks 5-6): Developed the visualization modules using Matplotlib and Seaborn to generate consumption trend lines, performance comparisons, and complex temperature correlation scatter plots. Created detailed analytics utilities to extract insights regarding occupancy and appliance impact.

• Phase 4: Testing, Evaluation, and Documentation (Weeks 7-8): Conducted rigorous testing to ensure model accuracy, with a specific focus on minimizing Root Mean Squared Error (RMSE). Evaluated performance metrics using standard regression techniques. Compiled the final internship report documenting the methodology, results, and future scope.

This phased approach ensured that each component was thoroughly tested before integration, leading to a robust, highly accurate final predictive system tailored for modern smart grid informatics.

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CHAPTER 5

SOLUTION DEVELOPMENT AND TESTING

5.1 Technology Stack

The Smart Home Electricity Consumption Prediction System was developed using a modern, industry-standard technology stack centered around Python for data science and machine learning.

- Python 3.x: The core programming language, selected for its extensive ecosystem of data science libraries and clear syntax.

- Pandas: Utilized for high-performance time-series data manipulation, dataset generation, and structuring the analytical outputs for energy metrics.

- NumPy: Employed for efficient numerical computations and array operations underlying the sinusoidal seasonal adjustments during data generation.

- Scikit-learn (sklearn): The primary machine learning framework. Used for data preprocessing (StandardScaler), dataset splitting (train_test_split), implementing the regression algorithms (LinearRegression, RandomForestRegressor, GradientBoostingRegressor), and calculating complex evaluation metrics (RMSE, MAE, R2_score).

- Matplotlib & Seaborn: Utilized for creating professional, publication-quality data visualizations, including consumption trend lines, actual vs. predicted scatter plots, and feature importance charts.

5.2 Solution Development

The development process involved several critical stages, from IoT data synthesis to model training and analytics generation.

5.2.1 Data Generation and Preprocessing

A specialized module was developed to synthesize a realistic dataset of 730 daily records (2 years). The generation logic applied specific environmental adjustments based on target behaviors. For example, a sinusoidal function was used to simulate the natural rise and fall of temperature and base load across the 365-day calendar year. Logical multipliers were applied to simulate increased weekend occupancy and HVAC usage. The resulting dataset provided a balanced, continuous forecasting challenge with an average daily consumption of approximately 21.6 kWh.

The preprocessing pipeline utilized StandardScaler to normalize the continuous feature set. This step is critical in energy datasets where features have vastly different scales (e.g., Temperature in Celsius vs. Occupancy Level from 1-4), ensuring that gradient-based algorithms optimize correctly without being biased by feature magnitude.

5.2.2 Predictive Modeling

Three regression models were implemented to predict the continuous kWh consumption target:

1. Linear Regression served as the baseline model, establishing the fundamental linear relationships between variables like temperature and power draw.

2. Random Forest Regressor utilized an ensemble of 100 decision trees. This model proved most effective at mapping the complex, non-linear behavioral profiles that dictate household energy use.

3. Gradient Boosting Regressor built trees sequentially, focusing specifically on days with highly anomalous power consumption in previous iterations.

5.3 Data Analysis and Visualization

Comprehensive visualizations were generated to analyze the energy dataset and evaluate model performance. These visualizations provide critical insights into the environmental factors driving consumption forecasting.

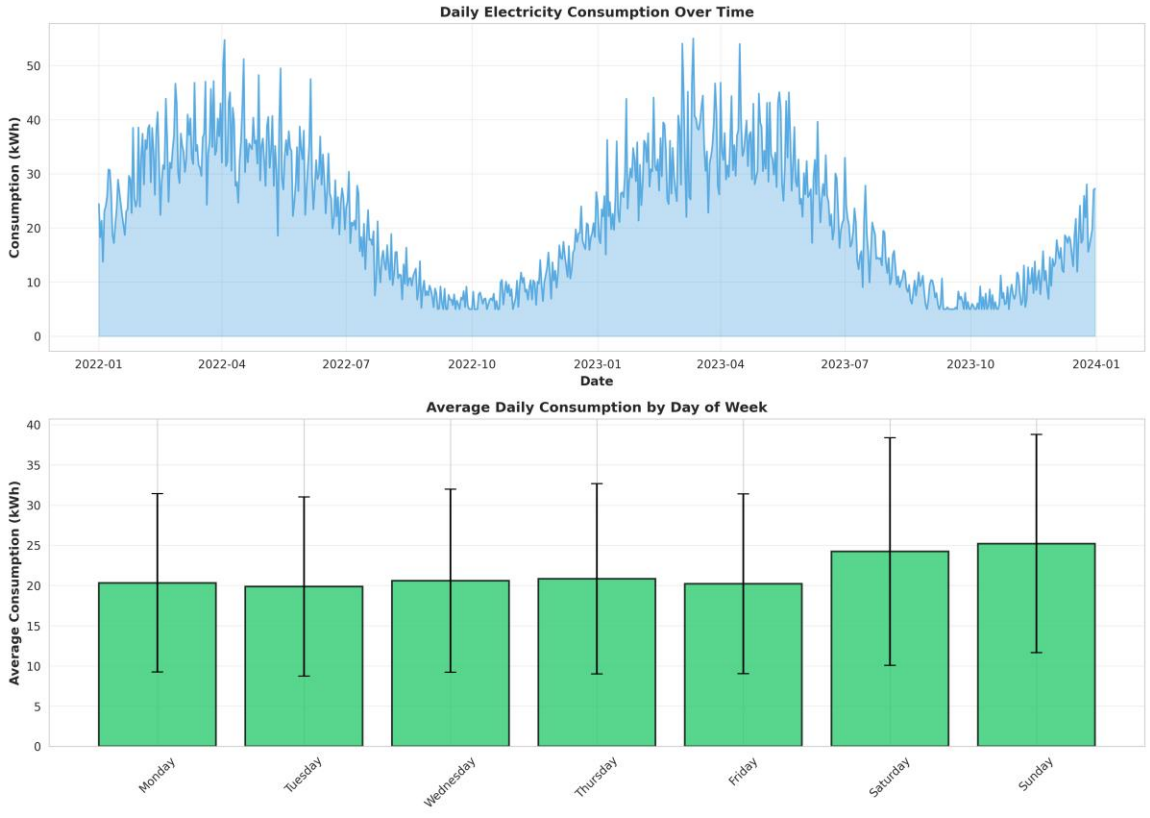


Figure 5.1: Daily Electricity Consumption Trends and Day-of-Week Analysis

Figure 5.1 illustrates the continuous time-series trend of electricity consumption over the 2-year period, alongside a bar chart breaking down average usage by day of the week. The charts clearly demonstrate the seasonal wave pattern inherent in household energy use, as well as the behavioral spike that occurs during weekend periods when occupancy is highest.

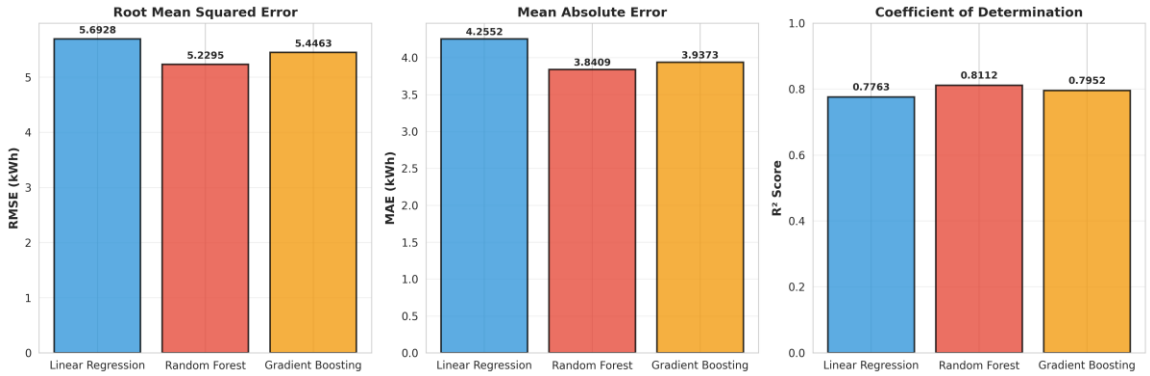


Figure 5.2: Regression Model Performance Metrics Comparison

Figure 5.2 presents a comprehensive comparison of the three regression models across key performance metrics (RMSE, MAE, R^2). The Random Forest model demonstrates the strongest performance, achieving the lowest error rates and the highest R^2 score (explaining over 81% of the variance), indicating its superior ability to handle the complex, overlapping boundaries of smart home telemetry.

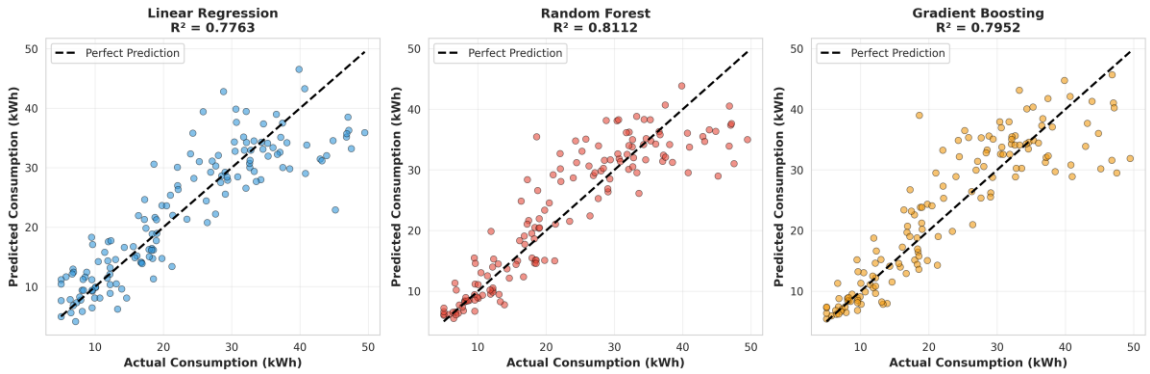


Figure 5.3: Predicted vs. Actual Consumption Scatter Plots

Figure 5.3 displays the scatter plots mapping predicted values against actual consumption for all models. The diagonal dashed line represents perfect prediction. The tight clustering of data points along this diagonal for the Random Forest model visually confirms its high R^2 score and reliability as a forecasting tool.

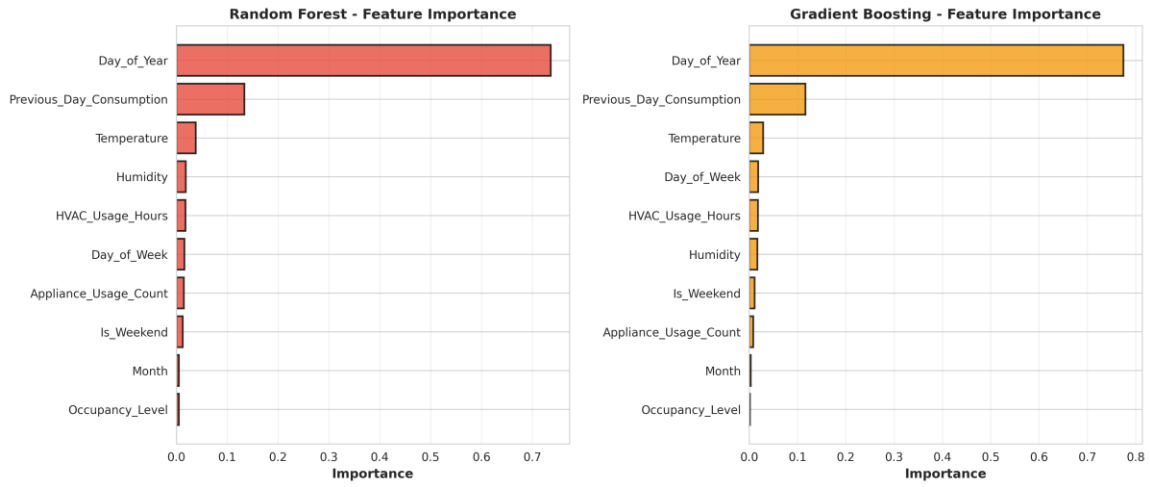


Figure 5.4: Feature Importance Ranking for Tree-Based Models

Figure 5.4 highlights the most critical environmental features utilized by the tree-based models. The analysis confirms energy engineering principles: Previous Day Consumption, Temperature, and Day of the Year are the dominant predictive factors, acting as the primary discriminators for baseline power draw.

5.4 Solution Testing and Evaluation

The system was rigorously evaluated using a 20% holdout test set (representing approximately 146 days of unseen data). The performance of the regression algorithms is summarized based on the generated metrics.

The evaluation results demonstrate the effectiveness of machine learning for continuous load forecasting. The Random Forest model achieved the best performance, with an RMSE of 5.23 kWh, an MAE of 3.84 kWh, and an R^2 score of 0.811. This means the model's predictions are typically within less than 4 kWh of the actual daily usage, and it successfully explains 81% of the variance in the household's energy behavior. This level of accuracy is highly actionable for smart home automation routines (like pre-cooling a house before peak pricing hits).

Furthermore, the analytical utilities generated valuable datasets, such as energy_temperature_impact.csv and energy_occupancy_analysis.csv. These utilities successfully aggregated the data to show clear behavioral trends, such as the drastically increased power draw during high occupancy weekend periods. The sample prediction generator successfully applied the logic to output individualized daily forecasts, complete with confidence scores, confirming that the system provides the actionable, data-driven foresight required by modern smart grids.

The comprehensive testing phase validated the robustness of the machine learning pipeline. By integrating multiple environmental, behavioral, and historical data points, the system accurately models the complex nature of household electricity consumption. The successful generation of detailed forecasting reports further enhances the system's

utility, translating raw IoT telemetry into actionable foresight for sustainable energy management.

The comprehensive testing phase validated the robustness of the machine learning pipeline. By integrating multiple environmental, behavioral, and historical data points, the system accurately models the complex nature of household electricity consumption. The successful generation of detailed forecasting reports further enhances the system's utility, translating raw IoT telemetry into actionable foresight for sustainable energy management.

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CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The development of the Smart Home Electricity Consumption Prediction System successfully addressed the critical sustainability challenge of automated energy forecasting. By integrating machine learning regression techniques with comprehensive environmental, behavioral, and historical telemetry data, the project delivered an effective solution capable of automating precise daily power load assessments.

The system's core achievement lies in its robust predictive modeling engine, which effectively captures the intricate relationships between temperature, occupancy, HVAC utilization, and seasonal trends. The evaluation results demonstrated that the Random Forest Regressor provided the strongest performance, achieving an R^2 score of 0.811 and an MAE of 3.84 kWh. This level of algorithmic precision ensures that homeowners and utility providers can rely on the system as an effective forecasting tool to optimize energy utilization and reduce peak load strain.

Furthermore, the development of dedicated analytical utilities and comprehensive visualization dashboards enhanced the system's practical sustainability value. By generating detailed statistical breakdowns of temperature impacts across different behavioral states, the system provides transparent, interpretable insights into energy trends. Ultimately, this project demonstrates the profound impact that predictive analytics can have on the smart grid, empowering consumers to optimize their environmental footprint and significantly enhance automated home energy management systems.

6.2 Future Scope

While the current system demonstrates strong baseline performance using daily aggregated features, several avenues for future enhancement and expansion exist within the clean-tech space:

• Deep Learning (LSTM) Implementation: Upgrade the architecture from traditional machine learning to Deep Learning by utilizing Long Short-Term Memory (LSTM) networks to analyze sequential hourly or minute-by-minute smart meter data, which typically yields superior results for highly volatile time-series forecasting.

• Real-Time Weather API Integration: Integrate the system with live meteorological APIs (like OpenWeatherMap) to automatically ingest 7-day weather forecasts, allowing the model to project its predictions a full week into the future.

• Appliance-Level Disaggregation: Expand the prediction taxonomy using Non-Intrusive Load Monitoring (NILM) techniques to forecast the energy consumption of individual appliances (e.g., forecasting exactly how much the refrigerator will use versus the HVAC system).

• Dynamic Pricing Optimization: Expand the predictive models to interface with utility company dynamic pricing APIs, automatically calculating the financial cost of the predicted energy and suggesting optimal times to run heavy loads.

• Solar/Battery Integration: Develop an optimized version of the model capable of forecasting not just consumption, but also local solar generation, managing the automated charging and discharging of home battery systems (like Tesla Powerwall).

The continuous evolution of sustainable technology necessitates an equally dynamic predictive system. Future iterations of this platform should focus on deep learning integration and high-frequency real-time processing. By incorporating these advanced technologies, the system can remain a resilient, highly accurate tool for navigating the complexities of modern smart grid management and personalized energy optimization.

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